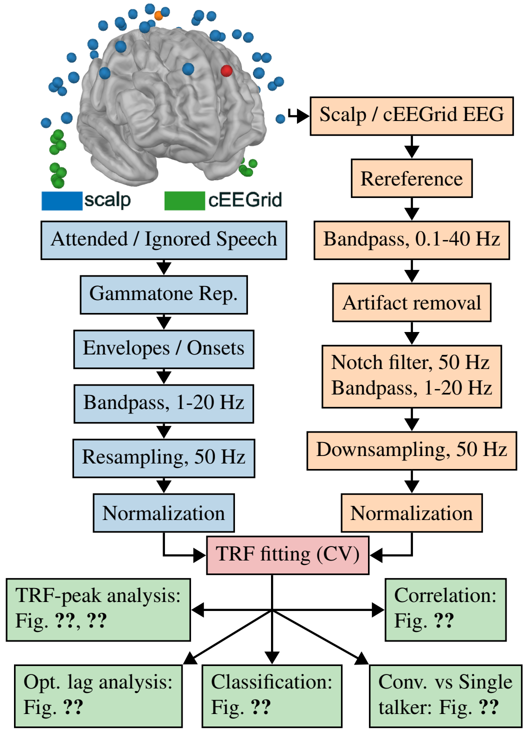

Figure 2: The top-left plot shows a three-dimensional rendering of electrode locations projected onto a standard brain model using MNE-Python. The plot illustrates the EEG setup, including the common mode sense (CMS) and driven right leg (DRL) electrodes in orange and red respectively. Scalp EEG electrodes are shown in blue, and cEEGrid electrodes are shown in green. The cEEGrid consists of 10 electrodes placed around each ear. In the rendering, the positions of cEEGrid arrays may not be exact. The subsequent preprocessing steps for the stimuli (light blue) and EEG data (light orange) are presented as flowcharts. Forward and backward temporal response functions (TRF) models are then fitted using cross-validation (CV) (red). The forward model is analyzed both qualitatively and quantitatively by inspecting TRF waveforms, while the backward model is analyzed quantitatively via correlation-based metrics, including reconstruction accuracy and classification performance (green).